

FYS191 All for One, One for All

An Exploration of Cooperative Behavior in the Animal Kingdom

Term: Fall 2023
Location: [107 Hasbrouck Laboratory Addition](#)
Time: Thurs 4:00 – 4:50 PM (16:00 – 16:50)
Credit Hours: 1 credit First Year Seminar
Prerequisites: None

Instructor: Seanne Clemente
Pronouns: he/they
[Why pronouns?](#)

Contact: srclemente@umass.edu
I will respond to emails within 24 hours. Although I may respond beyond normal working hours, I do not expect you to do the same.

Drop-in Hours: Weds 1:00 PM – 3:00 PM
In-person: [French Hall 205](#)
Zoom: ID: 585 876 0821 or [use this link](#)
[What are drop-in hours?](#)

Need to meet me, but can't make this time? Let me know, and we can arrange an alternative.

About this Course

Un pour tous – one for all! The benefits of working together are evident. Predators hunting in packs can effectively incapacitate and dispatch their prey. On the flip side, a large group of prey can stymie antagonists that are much more formidable than the group's component individuals. And in the utopian honey bees, every member works diligently for the sake of the hive – in some cases, even dying in the process.

Tous pour un – all for one! Yet, selfishness pervades even the most harmonious of groups. In the predator pack, who gets first dibs on a kill? In the herd of prey, who stays on the periphery, where a predator is most likely to strike? Even the fallen worker honey bee, who sacrifices everything in defense of her sisters, has something to gain from her actions.

Through the lens of evolution, life is simply a struggle between optimizing benefits and minimizing costs. In this course, we will explore why this balance favors cooperation in some species (and not in others). We will explore key examples of cooperative behaviors in animals, from temporary alliances to the so-called “superorganisms” that blur the line between individual and collective. Finally, we’ll take a close look at *Homo sapiens* – humans – and how they use cooperation as a powerful tool to shape the Earth to their benefit.

Class Materials

There are **no required texts for this course**, and any pertinent reading material will be provided to you on Canvas.

If you wish to delve deeper into the topics we discuss (and animal behavior in general), [Animal Behavior](#) by Dustin Rubinstein is an excellent resource to start. (Older versions of this book will be more economical; note that they are authored by John Alcock instead.)

If you found this class interesting, consider signing up for the full-fledged Animal Behavior course (BIO 550) and/or the Introductory Neurobiology course (BIO 372).

My Hopes for this Class

By the end of the course, I hope you can:

- Explain what fitness means from a biological perspective
- Give examples of cooperative behavior in animals
- Gain an increased appreciation of the natural world around you
- Identify resources on and off-campus that will help you succeed in college life and beyond

Remember, this is a one-credit course! As a rule of thumb, you should dedicate [3 hours per week](#) for each credit a class is worth. We meet for an hour, so try to spend no more than two hours per week on this course.



Classroom Community Culture

Because our course only meets once a week for 50 minutes, the vast majority of your time in this class will be spent on in-person activities and discussions.

I understand that group settings and open discussion may not be the most comfortable learning environment for everyone. To this end, I strongly encourage you to:

- Be your authentic self
- Advocate for yourself, and ask for help when you need it
- Ask questions when you are confused
- Share honest feedback with me, both good *and* bad

Each of you is an irreproducible combination of lived experiences and identity (e.g. culture, beliefs, and values). Further, recognize that your classmates may not have had the same resources and educational background that you have received to this point.

To further facilitate a learning environment where everyone feels welcome, I expect the following when you engage with your peers:

- **Above all, be kind to each other**
- Exercise patience and grace, *especially* when someone expresses an opinion that goes against your own
- Value everyone's contribution, and remember that speaking up can be exponentially more difficult for some people than others
- Speak as if someone in the room has the identity you are speaking about – this is often true
- Respect everyone's time by staying on topic

While I encourage diverse viewpoints, **I will not tolerate any harassment, aggression, or demeaning behavior towards your peers or myself.** If you feel that someone has acted inappropriately towards you during class, **please do not hesitate** to let me know.

Thank you for reading the syllabus up to this point. Search the Internet for an image of a red panda, and send me your favorite one! In exchange, you will receive credit for one assignment – use it at any point during the semester.

How Will You Be Graded?

Participation 60%

Assignments 20%

Development Project 20%

TOTAL **100%**

Assignments

All assignments will be submitted on Canvas. Examples of assignments include: watching a video and responding to a prompt, exploring the UMass campus, or practicing to write an email to your professor.

Participation

Most of your grade is related to participation. A large chunk of our time in class will involve group activities and open discussion. 1/3 of your participation grade is related to attendance (see *Attendance Policy*). You will determine the other 2/3rds of the grade: at the end of the semester, I will ask you to assess your own performance.

Development Project

At the end of the semester, you will be asked to work on an aspect of your development as a college student (and beyond). You will have the freedom to design your own project; otherwise, you may choose from several options such as: researching an on-campus lab you want to work in, creating a time management journal that tracks your activities for 1 week, crafting a factsheet for a few internships you want to apply to, drafting a 4-year plan.

Grading Scale

A	A-	B+	B	B-	C+	C	C-	D+	D	F
100 – 93	92.5 – 90	89.5 – 87	86.5 – 83	82.5 – 80	79.5 – 77	76.5 – 73	72.5 – 70	69.5 – 67	66.5 – 63	below 63

Attendance Policy

Because we only have 13 classes for the whole semester, missing even one meeting can significantly impact your engagement in this course. Attending *every* class is highly encouraged, but I understand that life is not predictable and situations will arise where you cannot make it.

You have one unexcused absence, no questions asked. For any further absence(s), I will need you **to inform me as soon as possible**. As long as you inform me of an absence beforehand, we can work out how to receive participation points for that day.



Deadlines and Late Work

All assignments will be due 2 weeks after they are given in class. Any submission after 11:59 PM Eastern on the due date is considered late. The late work policy is outlined below.

When did you submit it?	What is the highest grade you can get?
Before 11:59 PM on the due date	100%
Up to 1 week after the due date	70%
More than 1 week after the due date	0%

As with attendance, please contact me **as soon as possible** if you cannot turn in an assignment on time, and we can work out an alternative.

Semester Schedule

This schedule is subject to change. The online version of the syllabus will always reflect the most current schedule.

Week	Date	Topic
1	7 Sept	Starting Out: Introductions, How Does Science Work?
11 Sep		LAST DAY TO ADD/DROP CLASSES
2	14 Sept	The Economy of Working Together: Fitness, Costs & Benefits, and the Prisoner's Dilemma
3	21 Sept	
4	28 Sept	
5	5 Oct	Basic Survival: Finding Food and Defending Against Enemies
6	12 Oct	
7	19 Oct	Making Babies: Finding Mates and Parenting
8	26 Oct	
9	2 Nov	Et cetera: Other Types of Cooperation
10	9 Nov	The Superorganism: Altruism and Eusociality
11	16 Nov	
23 Nov		THANKSGIVING BREAK - NO CLASS
12	30 Nov	The Great Society: Special Topics on Cooperation in <i>Homo sapiens</i>
13	7 Dec	
21 Dec		Instructor must post grades before this date.

Accommodation

The University of Massachusetts Amherst is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with Disability Services (DS), you may be eligible for reasonable academic

accommodations to help you succeed in this course. If you have a documented disability that requires an accommodation, please notify me within the first two weeks of the semester so that we may make appropriate arrangements. For further information, please visit [Disability Services \(161 Whitmore Administration Building; ph: 413-545-0892\)](#).

Further, these resources on campus that may help:

- [Dean of Students Office](#)
 - [Writing Center](#)
 - [Learning Resource Center](#)
 - [Student Success](#)
 - [Center for Counseling and Psychological Health \(CCPH\)](#)
 - [English as a Second Language \(ESL\) Program](#)
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Academic Honesty

Since the integrity of the academic enterprise of any institution of higher education requires honesty in scholarship and research, academic honesty is required of all students at the University of Massachusetts Amherst. Academic dishonesty is prohibited in all programs of the University. Academic dishonesty includes but is not limited to: cheating, fabrication, plagiarism, and facilitating dishonesty. Appropriate sanctions may be imposed on any student who has committed an act of academic dishonesty. Instructors should take reasonable steps to address academic misconduct. Any person who has reason to believe that a student has committed academic dishonesty should bring such information to the attention of the appropriate course instructor as soon as possible. Instances of academic dishonesty not related to a specific course should be brought to the attention of the appropriate department Head or Chair. Since students are expected to be familiar with this policy and the commonly accepted standards of academic integrity, ignorance of such standards is not normally sufficient evidence of lack of intent. [More information on academic honesty here.](#)

Title IX

In accordance with Title IX of the Education Amendments of 1972 that prohibits gender-based discrimination in educational settings that receive federal funds, the University of Massachusetts Amherst is committed to providing a safe learning environment for all students, free from all forms of discrimination, including sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. This

includes interactions in person or online through digital platforms and social media. Title IX also protects against discrimination on the basis of pregnancy, childbirth, false pregnancy, miscarriage, abortion, or related conditions, including recovery. There are resources here on campus to support you. A summary of the available Title IX resources (confidential and non-confidential) [can be found here](#). You do not need to make a formal report to access them. If you need immediate support, you are not alone. Free and confidential support is available 24 hours a day / 7 days a week / 365 days a year at the SASA Hotline 413-545-0800.